Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Languages used to process data, perform calculations, and build the web application.	Python, HTML, CSS	Python supports the Flask application and application logic. HTML structures web pages, while CSS styles the website. Tableau calculated fields are used for analytical calculations.
2.	Database / Data Warehouse	Stores the structured healthcare data used for analysis.	CSV file (`smart_healthcare_dataset.csv`)	CSV provides a simple format for storing patient records and importing them into Tableau. A separate database or data warehouse is not required for the current project.
3.	Data Ingestion	Loads healthcare data into the analytical environment.	Tableau Desktop, CSV data connection	Tableau Desktop connects to the CSV dataset and makes its fields available for analysis and visualization.

4.	Data Transformation and Business Logic	Prepares data and calculates the metrics required for healthcare analysis.	Tableau calculated fields, filters, aggregations, Python (Flask application logic)	Tableau supports age-group and BMI-category calculations, patient counts, average blood pressure, and disease analysis. Python handles the web application's logic.
5.	Data Visualization / Analytics	Creates dashboards, charts, KPIs, and analytical reports.	Tableau Desktop, Tableau Public	Tableau Desktop is used to create interactive charts, KPI cards, dashboard filters, and the healthcare Story. Tableau Public publishes the dashboard and Story online.
6.	Front-End / User Interface	Provides the interface through which users view project information and access analytical results.	Flask, HTML, CSS, Tableau Public	Flask serves the website, HTML defines its page structure, and CSS provides styling. Tableau Public provides access to the published healthcare dashboards.
7.	Version Control / Project Repository	Stores and organizes source code, datasets, documentation, and project files.	GitHub	GitHub keeps the project files organized and provides a repository link for sharing and submission.
8.	Web Application Framework	Connects the website routes with the pages and project content.	Flask (Python)	Flask provides a lightweight framework for serving the Health360 website and rendering HTML templates.
9.	Development Environment	Supports writing, testing, and running the application.	Python, Flask development server, web browser, Tableau Desktop	These tools allow the website to be developed and tested locally and the analytics workbook to be created and checked.